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Apparatus and method to provide conditioning to a deposited powder

Abstract

An improved apparatus and method provide conditioning to a powder deposited on a substrate (e.g., a web), for example, by wetting the powder in a 3D printing apparatus. To achieve this in an exemplary implementation, a wetting apparatus is located between a powder dispenser and at least one top calendering roller in a direction of movement of the substrate, wherein the wetting apparatus is configured to apply a wetting agent to the powder on the substrate before the powder passes through the calendering roller. The wetting agent is comprised of a material which increases cohesiveness of the powder to prevent the powder from adhering to the top roller. In a particular implementation, the wetting agent is steam confined to an area of the substrate where the powder passes through the wetting apparatus, without wetting other areas of the substrate which are not in the wetting apparatus.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2837440	12/1957	Boivin	521/61	D06N 3/0052
3081269	12/1962	Shannon	521/181	C08J 9/0061
3402480	12/1967	Geitz	34/619	A24B 3/12
5876550	12/1998	Feygin et al.	N/A	N/A
6213168	12/2000	Gaylo	141/73	B29C 41/12
9475234	12/2015	Morikawa	N/A	B29C 64/35
2005/0104241	12/2004	Kritchman et al.	N/A	N/A
2006/0032840	12/2005	Bagavath-Singh	219/121.64	B23K 15/0086
2011/0190923	12/2010	Matsui et al.	N/A	N/A
2016/0243764	12/2015	Hays et al.	N/A	N/A
2020/0171752	12/2019	Rogren	N/A	N/A
2020/0307090	12/2019	Daniels et al.	N/A	N/A
2021/0008617	12/2020	Sachs et al.	N/A	N/A
2022/0032377	12/2021	Hudelson et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3461940	12/2018	EP	D04H 1/4266
3919260	12/2020	EP	N/A

OTHER PUBLICATIONS

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Background/Summary

TECHNICAL FIELD AND BACKGROUND

(1) The present disclosure pertains to an improved apparatus and method to provide conditioning to a powder deposited on a substrate by lubricating/wetting the powder, for example, in a 3D printing apparatus. Specifically, in current binder jetting 3D printing, after powder deposition on a substrate, the powder on the substrate may be passed through rollers, such as a calendering system, for compaction. Previous techniques suffer from non-uniform compaction due to the powder sticking to a contacting roller or splitting of the powder between the roller and a carrier plate or substrate on which the powder is deposited. In all cases, this results in non-uniform part fabrication.

Accordingly, there is a need for an arrangement for uniform compaction and preventing powder sticking and splitting during powder and pattern roller compaction, for example, in 3D printing, especially powder binder jetting (although the techniques described herein can also be applied to continuous material jetting systems). The practice of applying a lubricant and/or wetting agent to deposited powder to improve flowability of the powder, thereby aiding in compaction uniformity, and to minimize or control the tension between the substrate and the later-applied layers, has been discussed in U.S. application Ser. No. 17/375,781, which is hereby incorporated by reference.

SUMMARY OF THE INVENTION

(2) In an implementation, an apparatus is provided including a powder dispenser configured to dispense powder onto a substrate, a calender comprised of a top roller, located above the powder on the substrate, and a conditioning apparatus located between the powder dispenser and the calender in a direction of movement of the substrate, wherein the conditioning apparatus is configured to apply a conditioning agent to the powder on the substrate before the powder passes through the calender, wherein the conditioning agent is comprised of a material which increases cohesiveness of the powder.

(3) In another implementation, a method is provided including dispensing powder onto a substrate, applying a conditioning agent to the dispensed powder on the substrate before the powder passes through the calender, and compacting the powder, after the conditioning agent is applied, wherein the conditioning agent is comprised of a material which increases cohesiveness of the powder to provide for uniform compaction of the powder during compacting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawing figures depict one or more implementations in accord with the present teachings, by way of example only, not by way of limitation. In the figures, like reference numerals refer to the same or similar elements. Furthermore, it should be understood that the drawings are not necessarily to scale.

(2) FIG. 1 shows a powder conditioning, lubrication, or wetting, apparatus located above a

substrate in a 3D printing apparatus in accordance with aspects of the present disclosure.

(3) FIG. 2 shows another implementation of a powder conditioning, lubrication, or wetting, apparatus located above a substrate in a 3D printing apparatus in accordance with aspects of the present disclosure.

(4) FIG. 3 shows a cutaway view of elements of a wetting apparatus, in accordance with aspects of the present disclosure.

(5) FIG. 4 shows a perspective view of an elevation mechanism for lifting the wetting apparatus off of the substrate, in accordance with aspects of the present disclosure.

(6) FIG. 5 shows a perspective view of the wetting apparatus, including the use of a heatsink under the substrate, opposite the wetting apparatus above the substrate, in accordance with aspects of the present disclosure.

(7) FIG. 6 shows a cutaway end view of the wetting apparatus, including tilting a table the steaming head is mounted on to prevent possible water contamination in the steam chamber, in accordance with aspects of the present disclosure.

(8) FIG. 7 shows a system interface for controlling the generation and characteristics of the steam applied to a steaming head in the wetting apparatus, in accordance with aspects of the present disclosure.

(9) FIG. 8 is a block diagram showing an example computer system upon which aspects of this disclosure may be implemented.

DETAILED DESCRIPTION

(10) In the following detailed description, numerous specific details are set forth by way of examples to provide a thorough understanding of the disclosed subject matter. It may become apparent to persons of ordinary skill in the art, though, upon reading this disclosure, that one or more disclosed aspects may be practiced without such details. In addition, description of various example implementations according to this disclosure may include referencing of or to one or more known techniques or operations, and such referencing can be at relatively high-level, to avoid obscuring of various concepts, aspects and features thereof with details not particular to and not necessary for fully understanding the present disclosure.

(11) In the present disclosure, an arrangement is provided to pretreat or condition powder before a calendering operation to ensure uniform compaction and preventing powder sticking to the top roller and splitting during powder and pattern roller compaction. These arrangements are particularly useful, for example in a 3D printing apparatus, especially powder binder jetting (although the techniques described herein can also be applied to the continuous material jetting, or to other powder deposition systems).

(12) In particular, as described in a related provisional application by the Applicant (Application No. 63/299,867), entitled “Printing Method and Apparatus for Additive Manufacturing, Including In-situ Powder Regeneration,” filed on Jan. 14, 2022, and which is hereby incorporated by reference, a platform for AM manufacturing has been developed which is a modular base machine that builds each layer separately. Each individual layer passes several apparatuses, and each apparatus has a unique function. One of these apparatuses includes compaction of the powder in a calendering operation while the deposited powder layers are still in the green non-cured stage. Thus, a method and an apparatus are disclosed herein which are particularly adapted to the new platform described in the above-noted related application. Specifically, in the platform described in this related application, due to the nature of the platform, the powder layers can be conditioned individually by the techniques discussed in the present disclosure, so it is possible to improve the calendering of the layers in the platform described in the above-identified related application.

(13) The related application noted above describes a system and a method for applying powder to a substrate at a powder applying apparatus, removing first portions of the powder to create clean portions of the substrate between portions of deposited powder, moving the portions of deposited powder remaining on the substrate to a wetting apparatus and wetting the powder with a wetting

agent at the wetting apparatus, passing the wetted portions of deposited powder through calendering rollers to compact the powder, cutting the substrate at the clean portions to create substrate segments each respectively including one of the portions of the deposited powder, moving the substrate segments having the portions of deposited powder, to a binder applying apparatus and applying a binder to the powder at the binder applying apparatus to harden the deposited powder into layers, transferring the substrate segments having the layers mounted thereon to respective carrier plates, and moving the carrier plates, with the substrate segments having the layers mounted thereon, to a stacking apparatus, and inverting each of the carrier plates in order to separate the layers from the substrate segments and to stack the layers individually on one another to form a multilayer product. The present disclosure can be incorporated into the system described in the related application as part of the calendering of the powder while it is in a green, non-cured stage in the overall 3D printing operation. It is noted that, although a certain order of steps is discussed above, the order of the steps could be changed, if desired, without departing from the scope of this disclosure.

(14) FIG. 1 shows a powder conditioning, lubrication, or wetting, apparatus **10** located above a substrate in a 3D printing apparatus in accordance with aspects of the present disclosure. FIG. 2 shows another implementation of such a powder conditioning, lubrication, or wetting, apparatus utilizing an alternative arrangement for powder compaction. Referring to FIGS. 1 and 2, a spray or inkjet print head **12** is positioned after a powder dispenser **14** on a carrier substrate **16** (e.g., a carrier conveyor belt) relative to the travel direction to spray or inkjet print a conditioning agent **18**, for example a wetting agent, on powder **20** which has been deposited on the substrate **16** (noting that “substrate” is also referred to in the art as “web”) by the powder dispenser **14**. In the case of lubricating, a lubricating or wetting agent/liquid **18** with controlled surface tension, preferably with high surface tension, is deposited uniformly on the deposited powder **20**. Thus, the deposited powder **20** is wetted by the controlled wetting agent **18**. This wetting agent **18**, by providing enough lubrication, helps uniform compaction of the powder **20** and prevents splitting of the powder by increasing internal cohesion, increasing cohesiveness of the powder. As will be described below with regard to a particular implementation, the wetting agent **18** can be steam generated from water. However, other wetting agents could be used, including other materials added to water or organic materials to help improve uniform compaction and internal cohesion of the powder for subsequent processing. Also, in an alternative implementation, the conditioning agent **18** could be applied by evaporation.

(15) Referring to FIG. 1, to minimize the sticking or splitting of powder **20** during calendering, a controlled surface energy coating **28** can be applied to the surface of the top compaction roller **24** of the calender **22** (which also includes a lower compaction roller **26**) to prevent splitting and sticking the powder **20** to the top compaction roller **24**. FIG. 2 shows an alternative arrangement using a covering layer **28'**, such as a sheet rolled from a dispensing sheet roller, with a low surface energy applied to the top compaction roller **24** of the calender **22** or applied to another covering surface used during the calendering. The combination of the coating or the covering surface with controlled surface energy and a wetting agent **18** applied by the spray dispenser of the inkjet printer **12** prevents splitting and sticking the powder to the compaction roller **24**. The amount of wetting agent **18** applied to the powder can be less than 10 wt %.

(16) Still referring to FIGS. 1 and 2, after powder compaction by the calendering compaction rollers **24** and **26**, and before patterning, the carrier substrate **16** with the compacted powder **20** may undergo a drying process by a dryer **30** to remove and evaporate the remaining wetting agent **18** that was applied by spraying or ink jetting and that has not already evaporated. The dryer **30** could be an IR lamp, hot air, or it could be a pulse dryer for faster drying.

(17) Summarizing the above discussion of FIGS. 1 and 2 regarding the concept of single layer powder compaction, a dispenser **12** (e.g., the spray or inkjet print head) is provided for dispensing a conditioner, lubricant or wetting agent **18** to a layer of deposited powder **20**. This conditioning

agent, lubricant or wetting agent **18** can be deposited by using evaporation techniques, an inkjet, a spray or, as discussed below regarding a particular implementation, steam. The conditioning agent, for example, a lubricant/wetting agent **18** lubricates/wets the deposited powder **20** for uniform compaction and for preventing sticking and splitting, as discussed above. In addition, as discussed above, in supplement to the lubricant/wetting agent **18**, a coating can be applied to the top roller **24** of the pair of compaction rollers **24** and **26**, as shown in FIG. **1**, or a cover sheet **28** can be applied between the top roller **24** and an upper surface of the deposited powder **20**, as shown in FIG. **2**, to further enhance the prevention of sticking to the top roller **24** or the splitting of the powder **20** during compaction by the calendering roller(s) **24** and **26**. Also, it is noted that, although the implementations shown in FIGS. **1** and **2** show a pair of calendering rollers **24** and **26**, in an alternative implementation a single top roller **24** can be used to press the powder layer **20** between the top roller **24** and the substrate **16**. Alternatively, a solid support, such as a rigid plate (not shown), can be provided under the substrate **16** to operate in conjunction with a single top calendering roller **24** to compact the powder **20**.

(18) Referring to FIGS. **3-7**, all of these pertain to a particular implementation of the principles discussed above, specifically using steam as the lubricant or wetting agent **18** for the powder **20**. For convenience, in the following discussion the term wetting agent **18** will be used, rather the conditioner or lubricant, noting that the wetting of the powder **20** that results from the use of the wetting agent **18** has a specific conditioning effect, in this case a lubricating effect.

(19) With regard to using steam as the wetting agent **18**, in various experiments to implement wetting of the powder **20** prior to calendering, a variety of approaches were tried to find a wetting process that would not damage the dispersed powder **20**, and that would provide sufficient lubrication to improve flowability, but not so much that the wetting agent spreads beyond the desired powder boundaries. Experiments were run using sprayers and fine misters, with different nozzles and types of nozzles. Although using such sprayers and fine misters improved the compaction and uniformity of the powder **20** following compaction in the calendering rollers **24** and **26**, particularly when combined with coating **28** or covering **28'** of the top roller with a low surface energy material, as shown in FIGS. **1** and **2**, the results of some of these experiments were striations, pockets, and damage to the powder sample **20**. Specifically, it was determined from these experiments that it was difficult to control the area of spray or the amount of water (or other wetting agent **18**) applied to the surface of the powder **20**.

(20) Experiments were then run applying a steam form of water to the surface of the powder **20**. It was determined that this method eliminated the above-noted issues regarding striations, pockets, and damage to the powder sample. The steam created an even, controllable, method of powder wetting. In other words, the steam serves as the wetting agent in this implementation and will be referred to hereinafter as steam **18'**. The steam area was also very controllable using a steam chamber, as discussed below, so as to not wet the substrate area of powder **20** in portions of the substrate **16** outside the steam chamber. This eliminated powder damage in further processes downstream, following the powder compaction in the calendering rollers **24** and **26**.

(21) FIG. **3** shows a cutaway view of elements of a conditioning apparatus, in this example a wetting apparatus **10**. As can be seen in FIG. **3**, the moving substrate **16**, with a controlled thickness of powder **20** (typically controlled via a blade to level the powder, and/or devices such as rollers to promote uniformity of the deposited powder) enters the wetting apparatus **10**. The wetting apparatus **10** (e.g., a steam housing **32** comprising a steam chamber **34**) provides an isolated region within which generated steam **18'** can be confined, and through which the substrate **16**, with the powder **20** thereon, can pass. In other words, the steam **18'** is limited to the portion of the substrate **16** within the steam chamber **34**, leaving powder **20** on portions of the substrate **16** outside of the steam chamber **34** untouched by the steam **18'**.

(22) As illustrated in FIG. **3**, the wetting apparatus **10** may comprise a steam housing **32** which contains all elements required for the wetting process. The steam housing **32**, though optional, may

serve several purposes as described in greater detail below, one of them being providing a safety feature, protecting the user from unintended exposure from the generated steam **18'**. It may also provide a means of protecting all elements of the wetting process from unforeseen damage when the wetting apparatus **10** is moved away from the substrate **16**.

(23) Steam **18'** is initially generated using a steam generator (e.g., such as using an evaporator as shown in FIG. 7) that, in accordance with one implementation, will continue to produce steam **18'** until instructed not to. The steam generator can be selected to provide a minimum steam flow/rate for applying steam to a steam source input **38**, and the desired temperature range of the steam **18'**. The generated steam **18'** feeds into the steaming head **36** inside the steam housing **32** via the steam source input **38**. The steam **18'** travels down a heated tube (see FIG. 7) from the steam generator to the steam source input **38**, and then into the steaming head **36** (e.g., a heated steam dispensing wand). The steaming head **36** may comprise one or more steam dispensing heads or nozzles which dispense steam **18'** within the head **36**. The steaming head **36** can be a wide triangle shaped object that coats the incoming powder **20** on the substrate **16** with the forced steam **18'**. The dispensed steam **18'** wets the powder **20** as it passes under the heated steam dispenser (i.e., the steaming head **36**).

(24) The width of the steaming head **36** is preferably at least the width of powder **20** distributed on the substrate **16**, but slightly less than the width of the substrate **16** itself. Alternatively, the width of the steaming head **36** can be substantially the same width as the substrate **16**. In one embodiment, the steaming head **36** is configured so that the steam **18'** spreads out in a triangular fashion as it travels from the steam source input **38** to the powder **20** on the substrate **16**.

(25) In some implementations, a heater **40** can be attached to, or located adjacent to, the steam chamber **34** or, alternatively, to the steaming head **36** to help maintain the steam **18'** in the steam chamber **34** and/or the steaming head **36**. In some implementations, the heater **40** is shaped to conform to the shape of the steaming head **36**. In other implementations, the heater **40** can be shaped differently than the steaming head **36**, covering either a smaller area or a larger area than the width and height of the steaming head **36**. The heater **40** can include a heater core passing through it that generates heat that spreads throughout the rest of the heater **40**. A sensor (e.g., see FIG. 7) can be provided within the steam chamber **34** to activate the heater **40** as and when needed based on sensing the temperature and level of steam **18'** being dispersed by the steaming head **36**. In other implementations, the rate of steam production may be associated with the speed of the substrate **16** moving through the steam chamber **34**. Sensors and controls can be incorporated inside the steam chamber **34** to cause the rate of steam production or the speed of the substrate **16** to be modified as needed to ensure that an appropriate amount of steam **18'** is applied to the powder **20** so that it will remain intact during the subsequent compaction in the calendering rollers **24** and **26**. Dynamic modification of steam generation may be enabled based on data received from the various sensors being fed back to the controls.

(26) As shown in FIG. 3, the wetting apparatus **10** includes one or more vacuum apparatuses **42** within the steam housing **32**, preferably at least one of which is adjacent the steaming head **36**, that will remove excess vapor or any condensation that may occur outside the steam target area of the powder **20**. The steam target area is the powdered area of the substrate **16** within the wetting apparatus **10**. The vacuum apparatuses **42** may remove excess vapor or condensation out of the wetting apparatus **10**, and/or control the vapor pressure/flow rate within the steam chamber **34**. This in turn provides additional control of the moisture received by the deposited powder **20**. The vacuum apparatuses **42** may open to the surface of the powder **20**, but not all the way down to the substrate **16**. In some implementations, the location of the opening can be varied by the user, or automatically based on data acquired via sensors, as dictated by application. The vacuum apparatuses **42** may also incorporate hydrophilic material (not shown), for example felt strips, to aid in the extraction of condensed water vapor. The hydrophilic material may be continuously wetted or saturated with an aqueous solution to aid in operation. Moisture received by the powder

20 can be controlled by optimizing the various parameters of the wetting apparatus **10** (including the pressure provided by the vacuum apparatuses **42**) and the speed of the substrate **16** moving through the steam chamber **34**.

(27) As the substrate **16** traverses through the wetting apparatus **10**, excess moisture, if any, on the “pure” substrate portions (e.g., those areas of the substrate **16** where no powder **20** is supposed to be deposited) will most likely have evaporated, so no additional excess water removal operation, other than the vacuum apparatuses **42**, is required. However, in one implementation, as also shown in FIG. 3, a substrate cleaning actuator **44** can also be provided at the output of the wetting apparatus **10** to blow air onto the substrate **16** in areas where powder **20** is not deposited to clean off any excess powder that may have drifted onto those portions of the substrate **16**. In one configuration, the substrate clean actuator **44** may dispense air all the way to the surface of the substrate **16**.

(28) In another configuration, as shown in FIG. 3, in order to accommodate the moving substrate, the substrate clean actuator **44** may be actuated such that it is lowered to the surface of the substrate **16** only during times that it is not disposed above powdered areas and raised to be above the surface of the powder **20** at other times. The timing of the raising and lowering may be timed based on the speed of the substrate **16**, width of the powder segments, and width of the non-powdered segments. In other instances, the timing of the raising and lowering may be based on data received from sensors, such as image acquisition devices and the associated image processing required to determine the edges of powder, areas of powder and or the areas of “pure” substrate.

(29) In an alternative configuration, one or more sensors located on the steam housing **32** may register an edge of powder **20** as it exits the steam housing **32**. Once the edge has been registered, a cleaning tool (not shown), for example a flexible blade made of a soft “squeegee” style material, may be used to push and remove liquid from the surface of the substrate **16** where no powder **20** is supposed to be deposited. The tip of the squeegee style flexible blade is the only material to touch the substrate, so the material used for the flexible blade should be chosen as a material which will prevent damage to the substrate as the tip of the flexible blade scrapes over the substrate to remove powder from the substrate in the desired first portion. The material chosen for use as a flexible blade may also depend on the specific chemistry of the powder **20**, with a desire to select a blade material to which the powder does not adhere or react. In one implementation, based on the speed of the moving substrate **16**, the angle of approach of the blade may be adjusted, such that, as it traverses the substrate **16** from one side to the other, while the substrate **16** is moving, the cleaning tool traverses the substrate, traveling between two adjacent edges of powder **20** on the substrate, in a direction substantially orthogonal to the longitudinal sides of the substrate **16**.

(30) As shown in FIG. 4, hydrophilic material can be provided, for example felt strips **46** which run parallel to the edges of the substrate **16**. These felt strips **46** will trap condensation and allow it to be vacuumed up by the vacuum apparatuses **42**. Alternatives to felt may include other materials which are able to collect moisture, and which does not damage the powder or substrate **16**, materials such as sponge, “squeegee” material for example. In some implementations, the material strips may comprise two sides with each side addressing one or more different requirements. For example, one side providing for moisture collection and the other aiding in sealing.

(31) In one implementation, water vapor that condenses on the upper surface of the hydrophilic material (e.g. felt strips **46**), may be transferred via one or more ports **48** on a lower portion of the wetting apparatus **10**, to a water storage receptacle (not shown) for eventual reuse/recycling. From the ports **48**, the condensed water may travel through tubing to the water storage receptacle. In some instances, the user may select which of the one or more ports **48** is utilized, providing a further way in which to control the flow rate of steam in the steam chamber **34**, and enabling a uniform distribution of water vapor to be maintained. In other instances, which of the one or more ports **48** to utilize may be selected automatically, based on the data from sensors which identify the level or condensed water vapor extraction at each of the ports **48**. The hydrophilic material may be

continuously wetted or saturated with an aqueous solution to enable it to absorb and entrap condensed vapor, while permitting the water vapor to be drawn through the vacuum. In some implementations, at one or more locations, the hydrophilic material may extend from the upper surface to the ports **48**, providing a transfer mechanism for the condensed water to the ports **48**.

(32) Additional sensors may be incorporated to ensure that the wetting apparatus **10** is always provided with a source of steam **18'**, and does not run dry. Additional sensors may also be provided to ensure that the hydrophilic material (e.g. felt strips **46**) maintain a required saturation level to ensure that condensed water vapor can be extracted via the applied vacuum.

(33) In a further implementation, as also shown in FIG. **4**, the wetting apparatus **10** can be lifted using an elevation mechanism **50**, which raises a table on which the entire wetting apparatus **10** is located in an upwards direction to expose the felt strips **46** so that they can be replaced or cleaned if required, separating the wetting apparatus **10** from the substrate **16**.

(34) The elevation mechanism **50** may be operated manually, or electronically. In another implementation the wetting apparatus **10** may be hinged along one edge, such that it can be rotated to expose the felt strips **46** so that they can be replaced or cleaned if required. Rails or locations structures (not shown) can be provided either side of the substrate **16** providing a defined location for the wetting apparatus **10** to reside, such that when lowered down towards the substrate **16**, the edges of the wetting apparatus **10** align and seal against the felt **46**. The weight of the wetting apparatus **10** may additionally aid in facilitating sealing.

(35) Referring now to FIGS. **5** and **6**, a heat sink **52** may be provided beneath the substrate **16**, opposite the wetting apparatus **10** and the steaming head **36**, to disperse heat generated in the wetting apparatus **10** away from the bottom surface of the supporting substrate **16**. By virtue of this, the substrate **16**, which is typically formed of mylar, does not suffer any heat damage from the steaming operation. It is noted, of course, that other materials could be used for the substrate **16**. The dimensions of the heatsink **52** can be adjusted accordingly to ensure that the substrate **16** itself is not damaged by the steaming operation occurring within the steam housing **32** located directly above the heatsink **52**. The heatsink **52** also helps control the temperature of the powder **20** to ensure steam absorption and condensation.

(36) In one implementation, a fan **54** may be provided adjacent the heat sink **52**, to further enhance heat dissipation away from the heat sink **52**. FIG. **5** also shows the vacuum apparatuses **42** for condensed water removal and the substrate cleaning actuator **44**, shown in more detail in FIG. **3**.

(37) In another implementation, a heating device, such as heating pad **56**, may be disposed on the upper surface of the steam chamber **34** (see FIG. **3**) to reduce or at least minimize moisture condensing on the inner surface of the steam chamber **34**, above the substrate. The heating device **56** may be located on an outer surface of the steam chamber **34**, inner surface of the upper surface of the steam chamber **34**, or integrated therein.

(38) Referring to FIG. **6**, a mounting table **60** on which the steaming head **36** is mounted is configured to be tilted at an angle θ to ensure that the steam dispersed from the steaming head **36** does not condense and sit on a surface of the mounting table **60**, which could create possible water contamination. In one embodiment, the mounting table **60** is fixed at an angle θ , in another the mounting table **60** is configured such that the angle θ can be adjusted.

(39) Referring to FIG. **7**, a system interface **70** is shown to generate and control the steam used by the steaming head **36** in the steam housing **32** shown, for example, in FIGS. **3** and **4**. Specifically, the system interface **70** includes a precision pump **72** which receives water from a water inlet **74**, and which pumps a precise predetermined amount of the water to an evaporator **75**. The evaporator **75** boils the water from the pump **72** to generate steam **18'** and applies the steam to a steam line **76**, which, in turn, passes the steam **18'** to the steam source input **38** on the steaming head **36** within the steam housing **32**. As shown in FIG. **7**, optionally, the steam line **76** can be between 1-3 meters long and can be heated (to preserve the steam). In an alternative implementation, a mass flow controller (not shown) can be included after the evaporator **75** to meter and control the rate of

steam **18'** supplied to the steam chamber **34** by the evaporator **75**.

(40) Still referring to FIG. 7, in order to ensure that the appropriate amount and temperature of steam is generated, various sensors can be used. Specifically, a temperature sensor **77** can be provided between the evaporator **75** and the steam line **76** to ensure that the steam output of the evaporator **75** is sufficient. If the steam temperature is in excess of a predetermined appropriate temperature, then the pump **72** and the evaporator **75** can be controlled to decrease the steam temperature. Alternatively, if the temperature sensor **77** indicates that the steam temperature is too low, the pump **72** and/or evaporator **75** can be adjusted to increase the steam temperature. A display and a controller **78**, as shown in FIG. 7, allows an operator to control this. Alternatively, a feedback control system (not shown) can be used to perform this adjustment operation automatically. In addition, control of the pump **72** and evaporator **75** can also be affected based on an output of an optional external sensor **79** located in the steam housing **32**, which can measure temperature, volume and pressure of the steam within the steam housing **32**, as described above with reference to FIG. 3.

(41) The rate and temperature of steam generation controlled by the system interface **70** of FIG. 7, the temperature range of the heater **40** located on the steam chamber **34**/steaming head **36**, and the speed of the substrate **16**, for example, can all be selected based on the substrate **16** used, powder materials **20** deposited on the substrate **16**, etc. The aim is to ensure that the dispersed powder **20** is not damaged, and sufficient lubrication/wetting is provided to improve flowability, but not so much that the wetting agent **18** spreads beyond the desired powder boundaries. Additionally, the steam **18'** generated, and the lubrication/wetting provided should be selected such that a Mylar surface (such as a surface of a calendering roller **24**) placed on the lubricated/wetted powder deposition surface, does not "pick up" any of the deposited powder **20**, i.e., the powder does not adhere to it. To aid in the control of these parameters, additional sensors may be incorporated into the system. For example, sensors may be provided to measure the powder temperature, and a feedback control system (not shown) can be used to adjust the steam generation and/or the heatsink **52**, such that the steam generator (e.g., the evaporator **75**) and/or heat sink **52** may assist in controlling the powder temperature. Such additional sensors can monitor the powder temperature for condensation and absorption. The sensors can also provide information that can be used for automation of the manufacturing operations, if desired. The heat sink **52** may also have a water-cooling mechanism to control the temperature.

(42) FIG. 8 is a block diagram showing an example a computer system **800** upon which aspects of this disclosure, for example the display and controller **78** shown in FIG. 7, or a feedback control system, as mentioned above, may be implemented. The computer system **800** may include a bus **802** or other communication mechanism for communicating information, and a processor **804** coupled with the bus **802** for processing information. The computer system **800** may also include a main memory **806**, such as a random-access memory (RAM) or other dynamic storage device, coupled to the bus **802** for storing information and instructions to be executed by the processor **804**. The main memory **806** may also be used for storing temporary variables or other intermediate information during execution of instructions to be executed by the processor **804**. The computer system **800** may implement, for example, as the display and controller **78** shown in FIG. 7, of a feedback control system, as discussed above, to carry out the operations discussed above with reference to FIGS. 1-7.

(43) The computer system **800** may further include a read only memory (ROM) **808** or other static storage device coupled to the bus **802** for storing static information and instructions for the processor **804**. A storage device **810**, such as a flash or other non-volatile memory may be coupled to the bus **802** for storing information and instructions.

(44) The computer system **800** may be coupled via the bus **802** to a display **812**, such as a liquid crystal display (LCD), for displaying information. One or more user input devices, such as the example user input device **814** may be coupled to the bus **802**, and may be configured for receiving

various user inputs, such as user command selections and communicating these to the processor **804**, or to the main memory **806**. The user input device **814** may include physical structure, or virtual implementation, or both, providing user input modes or options, and a cursor control **816** for controlling, for example, a cursor, visible to a user through display **812** or through other techniques, and such modes or operations may include, for example virtual mouse, trackball, or cursor direction keys.

(45) The computer system **800** may include respective resources of the processor **804** executing, in an overlapping or interleaved manner, respective program instructions. Instructions may be read into the main memory **806** from another machine-readable medium, such as the storage device **810**. In some examples, hard-wired circuitry may be used in place of or in combination with software instructions. The term “machine-readable medium” as used herein refers to any medium that participates in providing data that causes a machine to operate in a specific fashion. Such a medium may take forms, including but not limited to, non-volatile media, volatile media, and transmission media. Non-volatile media may include, for example, optical or magnetic disks, such as storage device **810**. Transmission media may include optical paths, or electrical or acoustic signal propagation paths, and may include acoustic or light waves, such as those generated during radio-wave and infra-red data communications, that are capable of carrying instructions detectable by a physical mechanism for input to a machine.

(46) The computer system **800** may also include a communication interface **818** coupled to the bus **802**, for two-way data communication coupling to a network link **820** connected to a local network **822**. The network link **820** may provide data communication through one or more networks to other data devices. For example, the network link **820** may provide a connection through the local network **822** to a host computer **824** or to data equipment operated by an Internet Service Provider (ISP) **826** to access through the Internet **828** a server **830**, for example, to obtain code for an application program.

(47) It is noted that although the above description has primarily been presented in terms of wetting powder in the environment of 3D printing, the apparatus and method of this disclosure is not limited to only powder wetting for 3D printers, and they could be used in any situation where it is desired to wet powder to keep the powder intact for subsequent processing.

(48) It will be apparent that though the embodiment described and illustrated is for a substrate in the form of a moving substrate, such as a conveyor belt, the teaching disclosed herein may be applied to discrete substrate segments, or individual substrate segments.

(49) In the following, further features, characteristics and advantages of the invention will be described by means of items: Item 1. An apparatus including a powder dispenser configured to dispense powder onto a substrate, a calender comprised of a top roller, located above the powder on the substrate, a conditioning apparatus located between the powder dispenser and the calender in a direction of movement of the substrate, wherein the conditioning apparatus is configured to apply a conditioning agent to the powder on the substrate before the powder passes through the calender, wherein the conditioning agent is comprised of a material which increases cohesiveness of the powder. Item 2. The apparatus of item 1, wherein the top roller has a controlled surface energy coating configured to prevent the powder from adhering to the top roller. Item 3. The apparatus of item 1 or 2, further comprising a covering layer located between the top roller and an upper surface of the powder on the substrate and configured to prevent the powder from adhering to the top roller. Item 4. The apparatus of any one of items 1-3, further comprising a dryer, located after the calender along the direction of movement of the substrate, the dryer being configured to dry the powder on the substrate after it has passed through the calender. Item 5. The apparatus of any one of items 1-4, wherein the conditioning apparatus comprises a wetting apparatus, and the wetting apparatus comprising a steaming head configured to apply steam to the powder on the substrate as a wetting agent. Item 6. The apparatus of any one of items 1-5, wherein the steaming head is located in a steam housing comprising a steam chamber covering a portion of the substrate along the direction

of travel of the substrate, the steam chamber being configured to confine steam from the steaming head to the portion of the substrate located within the steam chamber. Item 7. The apparatus of any one of items 1-6, further comprising at least one vacuum apparatus located in the steam housing of the wetting apparatus, adjacent to the steaming head, and configured to remove excess steam and condensation generated by the steaming head in the steam housing. Item 8. The apparatus of any one of items 1-7, wherein the wetting apparatus is configured to continuously generate steam at the steaming head while the substrate moves through the wetting apparatus. Item 9. The apparatus of any one of items 1-8, further comprising a sensor for sensing when powder is on a portion of the substrate that is under the steaming head, wherein the wetting apparatus is configured to only generate steam from the steaming head when the sensor determines that there is powder on the portion of the substrate under the steaming head. Item 10. The apparatus of any one of items 1-9, further comprising a heater located adjacent to the steaming head and configured to maintain steam in the steaming head. Item 11. The apparatus of any one of items 1-10, further comprising an elevation mechanism configured to separate the wetting apparatus from the substrate. Item 12. The apparatus of any one of items 1-11, further comprising a heat sink located under the substrate opposite the steam housing of the wetting apparatus. Item 13. The apparatus of any one of items 1-12, further comprising a sensor configured to measure powder temperature. Item 14. The apparatus of any one of items 1-13, further comprising a sensor configured to measure temperature of steam as it enters the steam chamber. Item 15. The apparatus of any one of items 1-14, wherein the heat sink is configured to have a controllable heat capacity to control powder temperature. Item 16. The apparatus of any one of items 1-15, further comprising a bottom roller located under the substrate, opposite to the top roller, so that the substrate is located between the top and bottom roller. Item 17. The apparatus of any one of items 1-16, wherein the substrate is comprised of a conveyor belt. Item 18. The apparatus of any one of items 1-17, further comprising a substrate cleaning actuator configured to blow air onto the substrate in areas where powder is not deposited to clean off any excess powder that may have drifted onto those portions of the substrate. Item 19. The apparatus of any one of items 1-18, wherein the substrate cleaning actuator is configured to be lowered and raised to come into and out of contact with the substrate. Item 20. A method including dispensing powder onto a substrate, applying a conditioning agent to the dispensed powder on the substrate before the powder passes through the calender, and compacting the powder, after the conditioning agent is applied, wherein the conditioning agent is comprised of a material which increases cohesiveness of the powder to provide for uniform compaction of the powder during compacting. Item 21: The method of item 20, further comprising drying the powder after compacting the powder. Item 22: The method of item 20 or 21, wherein the conditioning agent is steam applied by a steaming head. Item 23: The method of any one of items 20-22, wherein the steaming head is located in a steam housing comprising a steam chamber covering a portion of the substrate along the direction of travel of the substrate. Item 24: The method of any one of items 20-23, further comprising applying a vacuum by at least one vacuum apparatus located in the steam housing, adjacent to the steaming head, to remove excess steam and condensation generated by the steaming head in the steam housing. Item 25: The method of any one of items 20-24, further comprising continuously generating steam at the steaming head while the substrate moves through the wetting apparatus. Item 26: The method of any one of items 20-25, further comprising sensing, via a sensor, when powder is on a portion of the substrate that is under the steaming head, wherein steam is only generated from the steaming head when the sensor determines that there is powder on the portion of the substrate under the steaming head. Item 27: The method of any one of items 20-26, further comprising heating the steaming head to maintain steam in the steaming head. Item 28: The method of any one of items 20-27, further comprising elevating the steaming head to separate the wetting apparatus from the substrate. Item 29: The method of any one of items 20-28, further comprising removing heat from the substrate via a heat sink located under the substrate opposite the steam housing. Item 30: The method of any one of items 20-29, further comprising measuring

temperature of the powder by a sensor. Item 31: The method of any one of items 20-30, further comprising measuring temperature of steam as it enters the steam chamber. Item 32: The method of any one of items 20-31, further comprising controlling powder temperature via a heat sink having a controllable heat capacity. Item 33: The method of any one of items 20-32, wherein compacting the powder is performed by passing the powder through a calender including a top roller above the substrate and a bottom roller located under the substrate, opposite to the top roller. Item 34: The method of any one of items 20-33, further comprising blowing air onto the substrate with a substrate cleaning actuator in areas where powder is not deposited to clean off any excess powder that may have drifted onto those portions of the substrate. Item 35: The method of any one of items 20-34, further comprising raising and lowering the substrate cleaning actuator to come into and out of contact with the substrate.

(50) While various embodiments have been described, the description is intended to be exemplary, rather than limiting, and it is understood that many more embodiments and implementations are possible that are within the scope of the embodiments. Although many possible combinations of features are shown in the accompanying figures and discussed in this detailed description, many other combinations of the disclosed features are possible. Any feature of any embodiment may be used in combination with or substituted for any other feature or element in any other embodiment unless specifically restricted. Therefore, it will be understood that any of the features shown and/or discussed in the present disclosure may be implemented together in any suitable combination.

(51) While the foregoing has described what are considered to be the best mode and/or other examples, it is understood that various modifications may be made therein and that the subject matter disclosed herein may be implemented in various forms and examples, and that the teachings may be applied in numerous applications, only some of which have been described herein.

(52) Unless otherwise stated, all measurements, values, ratings, positions, magnitudes, sizes, and other specifications that are set forth in this specification are approximate, not exact. They are intended to have a reasonable range that is consistent with the functions to which they relate and with what is customary in the art to which they pertain.

(53) It will be understood that the terms and expressions used herein have the ordinary meaning as is accorded to such terms and expressions with respect to their corresponding respective areas of inquiry and study except where specific meanings have otherwise been set forth herein. Relational terms such as first and second and the like may be used solely to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “a” or “an” does not, without further constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

Claims

1. An apparatus, comprising: a powder dispenser, the powder dispenser configured to dispense powder onto a substrate; a blade for controlling the thickness of the powder on the substrate and promoting uniformity of the powder; a calender, the calender comprising a top roller, the top roller configured to be located above the powder on the substrate; a wetting apparatus, the wetting apparatus configured to be located between the powder dispenser and the calender, the wetting apparatus comprising a steam head, the steam head configured to apply steam to the powder prior to the powder passing through the calender; a sensor for sensing when powder is on a portion of the substrate under the steam head; a vacuum positioned adjacent to the steam head and configured to

control the vapor flow rate to control the moisture received by the dispensed powder; and wherein the wetting apparatus is configured to generate steam based on a determination of the powder being on a portion of the substrate under the steam head; and wherein the substrate comprises a conveyor belt, the conveyor belt configured to move the powder through the wetting apparatus and the calender.

2. The apparatus of claim 1, wherein the top roller comprises a controlled surface energy coating configured to prevent the powder from adhering to the top roller.
3. The apparatus of claim 1, further comprising a covering layer positioned between the top roller and an upper surface of the powder on the substrate, the covering layer configured to prevent the powder from adhering to the top roller.
4. The apparatus of claim 3, further comprising a dryer configured to dry the powder on the substrate.
5. The apparatus of claim 1, wherein the steam head is located in a steam housing, the steam housing comprising a steam chamber covering a portion of the substrate.
6. The apparatus of claim 5, wherein the vacuum is positioned inside the steam housing and further configured to remove excess steam and condensation generated by the steam head in the steam housing.
7. The apparatus of claim 1, wherein the wetting apparatus is configured to continuously generate steam at the steam head while the substrate moves through the wetting apparatus.
8. The apparatus of claim 1, wherein the wetting apparatus is configured to only generate steam from the steam head when the sensor determines that there is powder on the portion of the substrate under the steam head.
9. The apparatus of claim 1, further comprising a heater located adjacent to the steam head and configured to maintain steam in the steam head.
10. The apparatus of claim 1, further comprising an elevation mechanism, the elevation mechanism configured to separate the wetting apparatus from the substrate.
11. The apparatus of claim 5, further comprising a heat sink, the heat sink being located under the substrate and opposite the steam housing of the wetting apparatus.
12. The apparatus of claim 1, further comprising a sensor, the sensor being configured to measure powder temperature.
13. The apparatus of claim 5, further comprising a sensor, the sensor being configured to measure temperature of steam as it enters the steam chamber.
14. The apparatus of claim 11, wherein the heat sink is configured to have a controllable heat capacity to control powder temperature.
15. The apparatus of claim 1, further comprising a bottom roller, the bottom roller configured to be located under the substrate and opposite to the top roller so that the substrate is located between the top roller and the bottom roller.
16. The apparatus of claim 1, wherein the substrate comprises one or more segments containing dispensed powder.
17. The apparatus of claim 1, further comprising a substrate cleaning actuator, the substrate cleaning actuator configured to blow air onto the substrate in areas where powder is not to be deposited to clean off excess powder.
18. The apparatus of claim 17, wherein the substrate cleaning actuator is configured to be lowered and raised to come into and out of contact with the substrate.
19. An apparatus, comprising: a powder dispenser, the powder dispenser configured to dispense powder onto a substrate, the substrate comprising one or more segments for receiving powder dispensed by the powder dispenser; a blade for controlling the thickness of the powder on the substrate and promoting uniformity of the powder; a calender, the calender comprising a top roller and a bottom roller, the top roller being located above the powder dispensed on the substrate, and the bottom roller being located under the substrate and opposite to the top roller, the substrate being

located between the top roller and the bottom roller; wherein the top roller comprises a controlled surface energy coating configured to contact the dispensed powder and prevent the dispensed powder from adhering to the top roller; a conditioning apparatus, the conditioning apparatus being located between the powder dispenser and the calender, wherein the conditioning apparatus is configured to apply a conditioning agent to the powder on the substrate before the powder passes through the calender; a sensor for sensing when powder is on a portion of the substrate under the conditioning apparatus; and a vacuum positioned adjacent to the conditioning apparatus and configured to control the conditioning agent flow rate to control the amount of conditioning agent received by the dispensed powder; and wherein the conditioning agent comprises a material that increases the cohesiveness of the powder.

20. An apparatus, comprising: a powder dispenser, the powder dispenser configured to dispense powder onto a substrate, wherein the substrate comprises a conveyor belt; a calender, the calender comprising a top roller, the top roller configured to be located above the powder on the substrate; a blade for controlling the thickness of the dispensed powder on the substrate and promoting uniformity of the dispensed powder; a conditioning apparatus, the conditioning apparatus being located between the powder dispenser and the calender, wherein the conditioning apparatus is configured to apply a conditioning agent to the powder on the substrate before the powder passes through the calender; a sensor for sensing when powder is on a portion of the substrate under the conditioning apparatus; and a vacuum positioned adjacent to the conditioning apparatus and configured to control the conditioning agent flow rate to control the amount of conditioning agent received by the dispensed powder; and wherein the conditioning agent comprises a material that increases the cohesiveness of the powder; and wherein the conveyor belt is configured to move the powder through the conditioning apparatus and the calender to increase the cohesiveness of the powder then compact the powder.
